GEOGRAPHY

INDIAN GEOGRAPHY

INDIAN CLIMATE

MODULE - 4

INDIAN MONSOON

- THE RHYTHM OF SEASONS

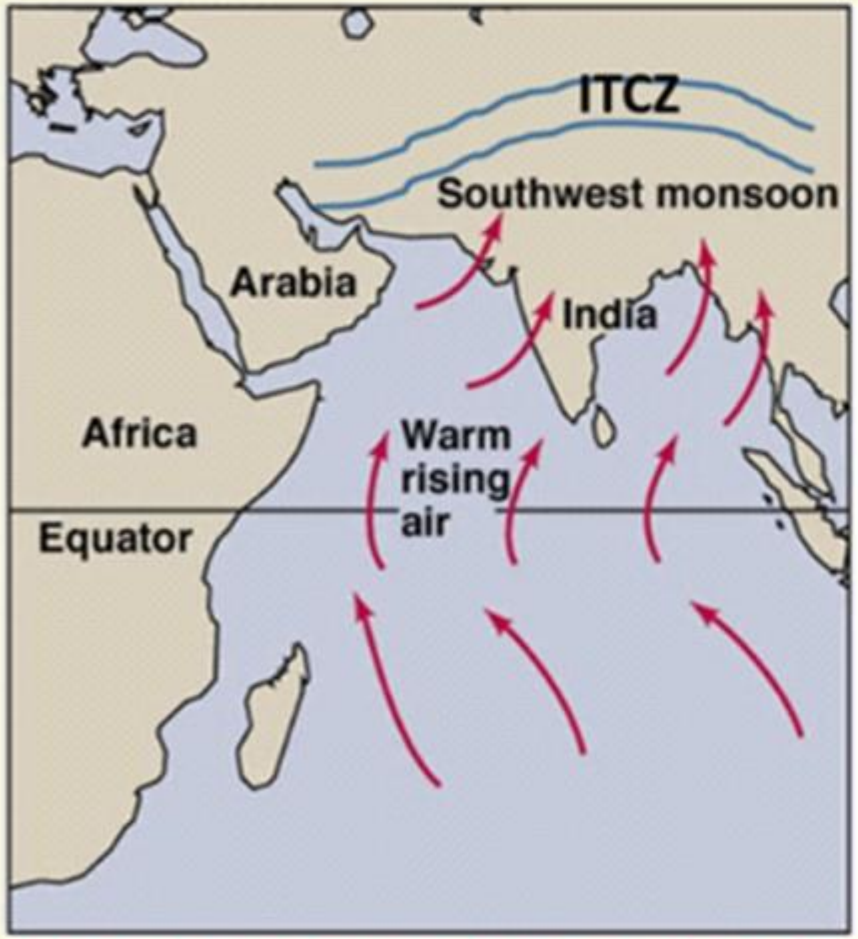
THE NATURE OF INDIAN MONSOON

Learning Space

1. Monsoon is a familiar though a little known climatic phenomenon.
2. Despite the observations spread over centuries, the monsoon continues to puzzle the scientists.
3. Many attempts have been made to discover the exact nature and causation of monsoon, but so far, no single theory has been able to explain the monsoon fully.
4. A real breakthrough has come recently when it was studied at the global rather than at regional level.
5. Systematic studies of the causes of rainfall in the South Asian region help to understand the causes and salient features of the monsoon, particularly some of its important aspects, such as:
   i. The onset of the monsoon.
   ii. Rain-bearing systems (e.g. tropical cyclones) and the relationship between their frequency and distribution of monsoon rainfall.
   iii. Break in the monsoon.

THE NATURE OF INDIAN MONSOON

1 ONSET OF THE MONSOON



HADLEY'S THERMAL THEORY

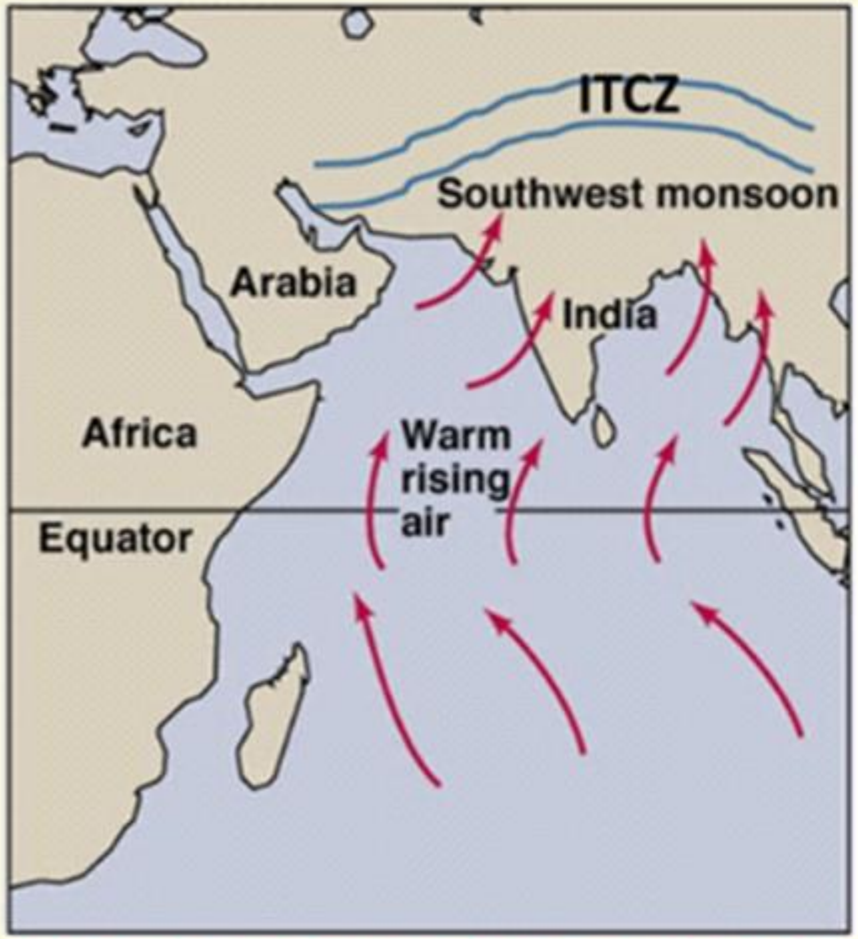
1. Towards the end of the nineteenth century, it was believed that the differential heating of land and sea during the summer months is the mechanism which sets the stage for the monsoon winds to drift towards the subcontinent.

2. During April and May when the sun shines vertically over the Tropic of Cancer, the large landmass in the north of Indian Ocean gets intensely heated.

3. This causes the formation of an intense low pressure in the northwestern part of the subcontinent.

THE NATURE OF INDIAN MONSOON

1 ONSET OF THE MONSOON



HADLEY'S THERMAL THEORY

4. Since the pressure in the Indian Ocean in the south of the landmass is high as water gets heated slowly, the low pressure cell attracts the southeast trades across the Equator.

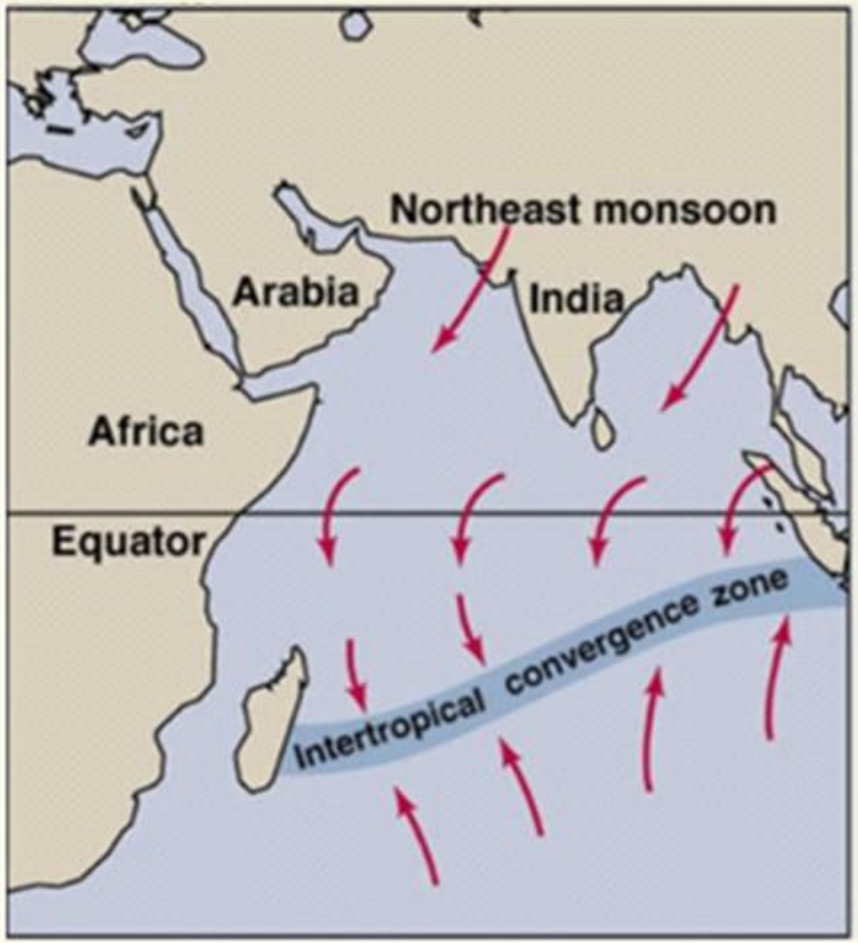
5. These conditions help in the northward shift in the position of the ITCZ.

6. The southwest monsoon may thus, be seen as a continuation of the southeast trades deflected towards the Indian subcontinent after crossing the Equator.

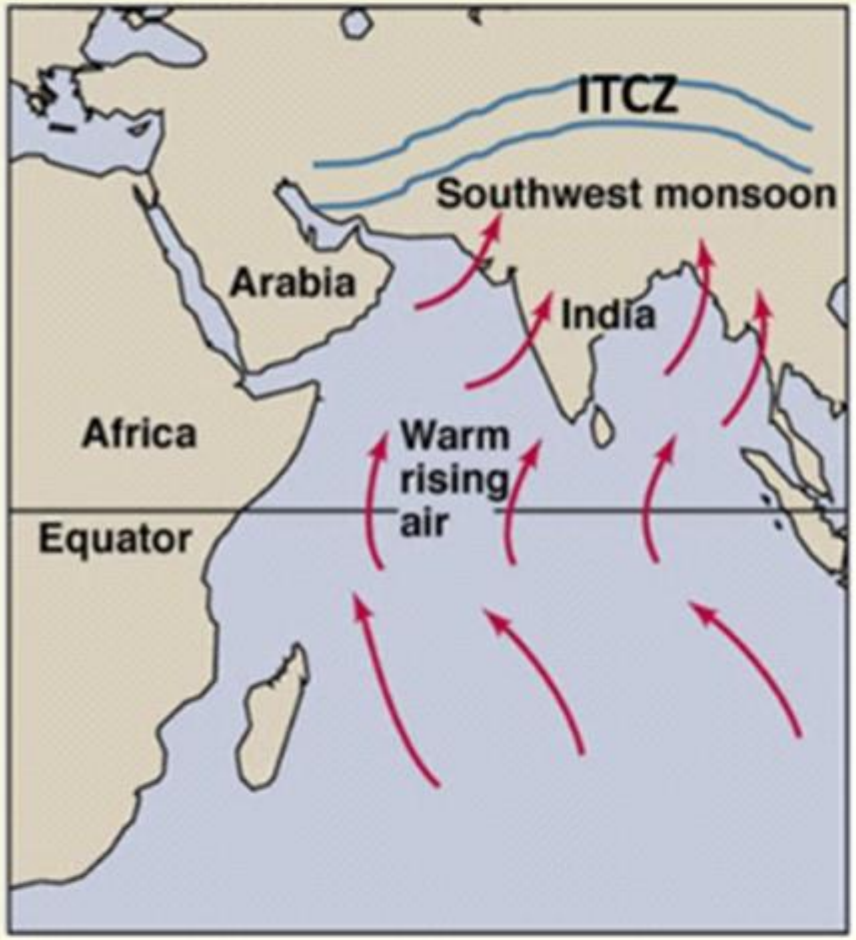
THE NATURE OF INDIAN MONSOON

1 ONSET OF THE MONSOON

HADLEY'S THERMAL THEORY



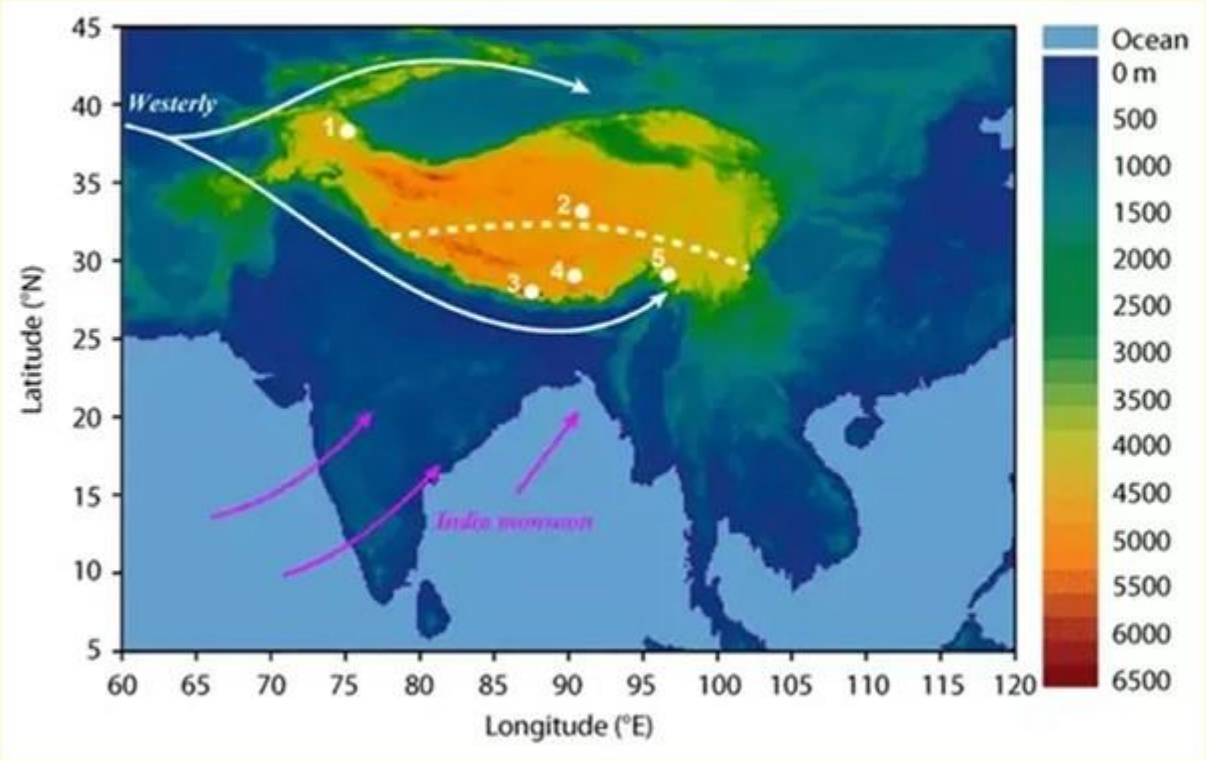
NORTH-EAST MONSOON



SOUTH-WEST MONSOON

THE NATURE OF INDIAN MONSOON

1 ONSET OF THE MONSOON



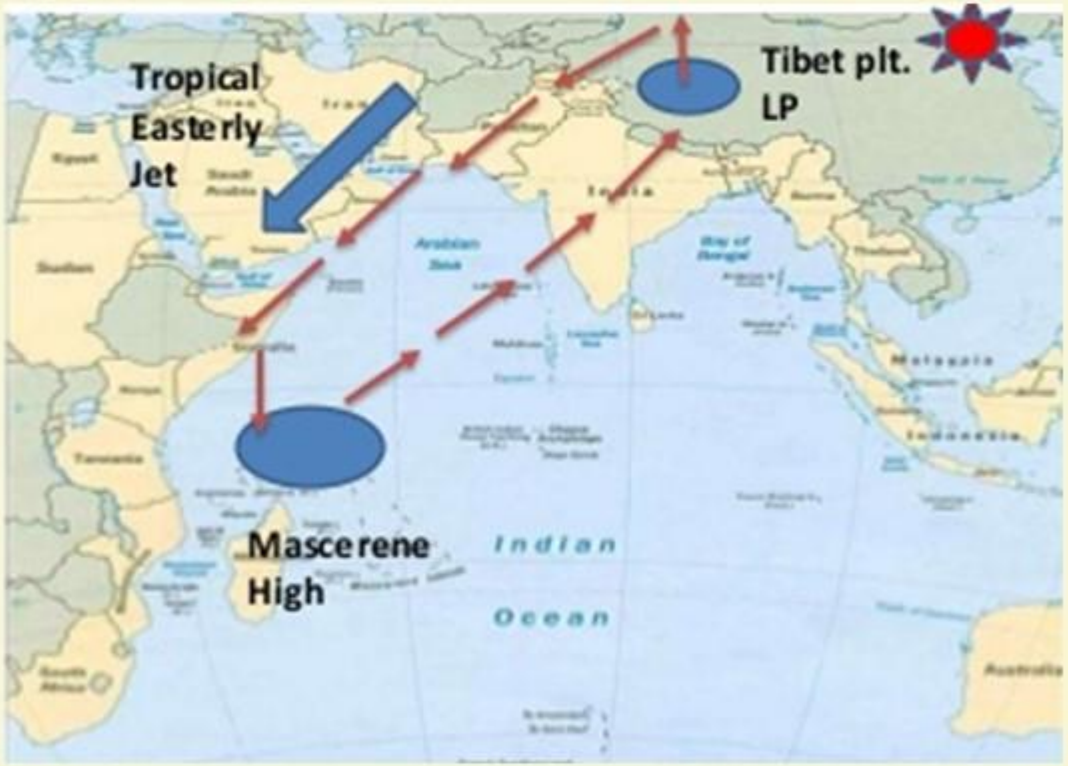
WITHDRAWAL OF STWJ AID IN THE DEVELOPMENT OF MONSOON

7. These winds cross the Equator between 40°E and 60°E longitudes.

8. The shift in the position of the ITCZ is also related to the phenomenon of the withdrawal of the westerly jet stream from its position over the north Indian plain, south of the Himalayas.

THE NATURE OF INDIAN MONSOON

1 ONSET OF THE MONSOON



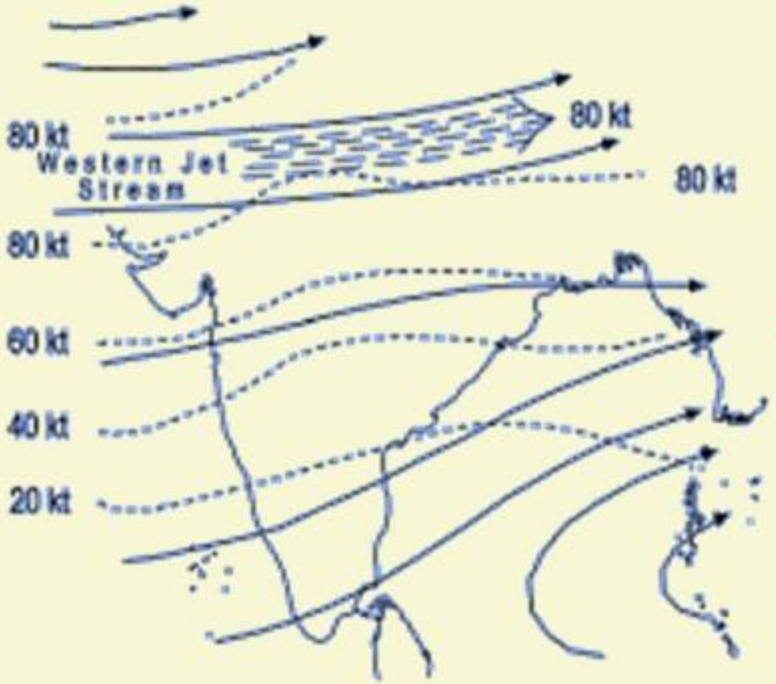
9 The easterly jet stream sets in along 15°N latitude only after the western jet stream has withdrawn itself from the region.

10 This easterly jet stream is held responsible for the burst of the monsoon in India.

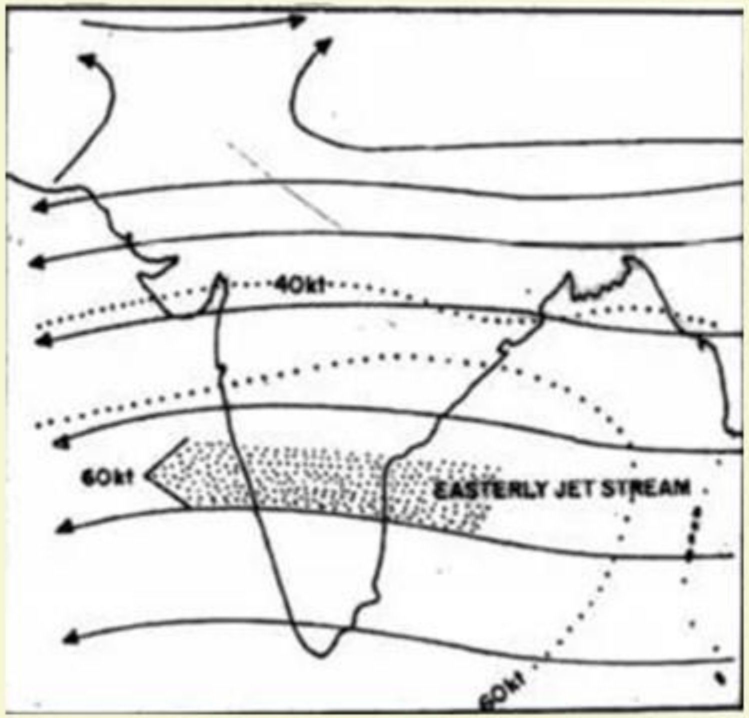
THE NATURE OF INDIAN MONSOON

1 ONSET OF THE MONSOON

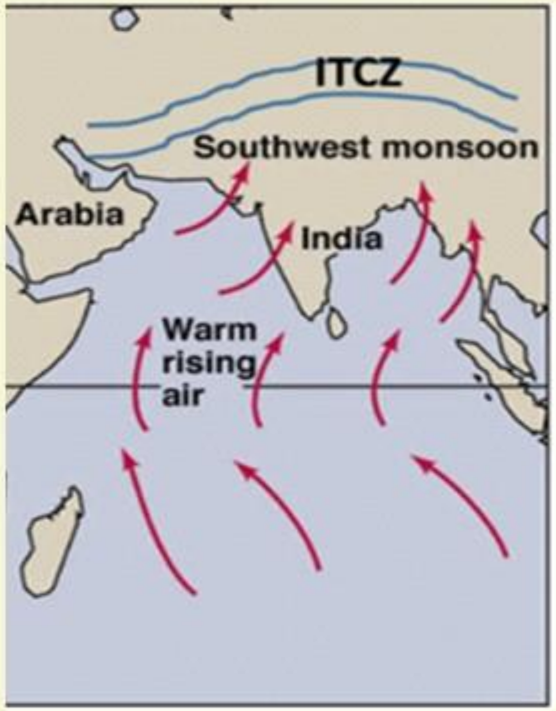
SEQUENCE OF EVENTS



Withdrawal of Westerly Jetstream



Establishment of Easterly Jetstream



Push of SW Monsoon to India

Learning Space

THE NATURE OF INDIAN MONSOON

1 ONSET OF THE MONSOON

ENTRY OF MONSOON INTO INDIA

1. The southwest monsoon sets in over the Kerala coast by 1st June and moves swiftly to reach Mumbai and Kolkata between 10th and 13th June.
2. By mid-July, southwest monsoon engulfs the entire subcontinent.

Learning Space

THE NATURE OF INDIAN MONSOON

1 ONSET OF THE MONSOON

RAIN-BEARING SYSTEMS AND RAINFALL DISTRIBUTION

1. There seem to be two rain-bearing systems in India.
2. First originate in the Bay of Bengal causing rainfall over the plains of north India.
3. Second is the Arabian Sea current of the south-west monsoon which brings rain to the west coast of India.
4. Much of the rainfall along the Western Ghats is orographic as the moist air is obstructed and forced to rise along the Ghats.
5. The frequency of the tropical depressions originating from the Bay of Bengal varies from year to year.
6. Their paths over India are mainly determined by the position of ITCZ which is generally termed as the monsoon trough.

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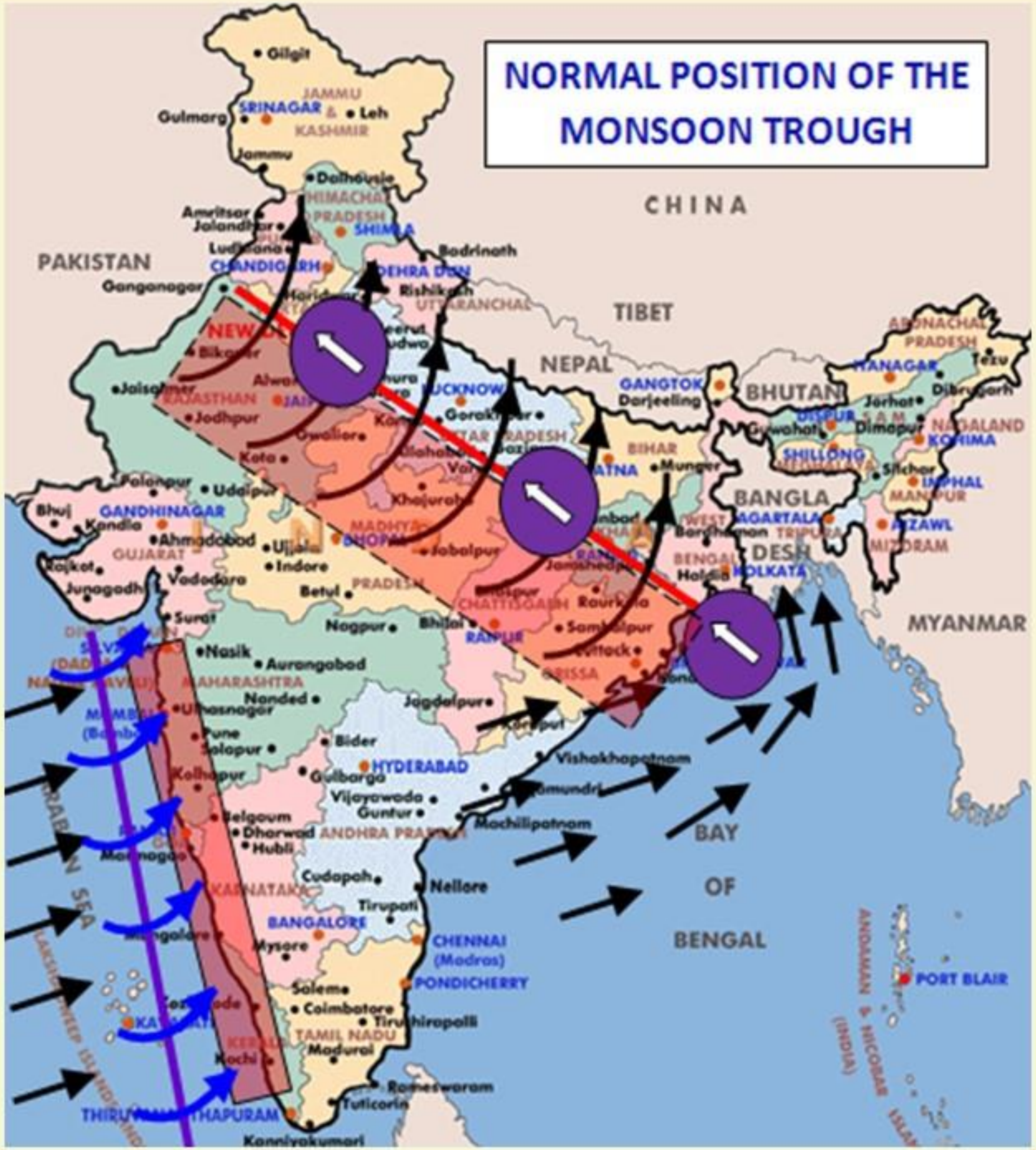
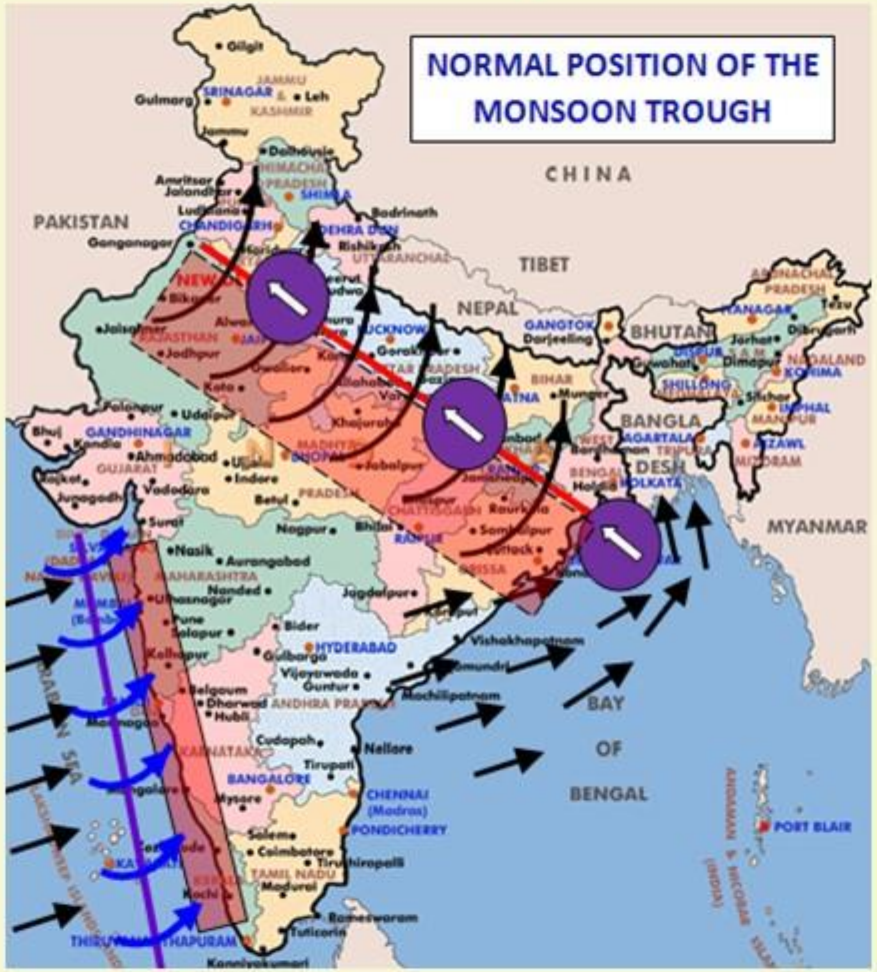


THE NATURE OF INDIAN MONSOON

1 ONSET OF THE MONSOON

RAIN-BEARING SYSTEMS AND RAINFALL DISTRIBUTION



8 As the axis of the monsoon trough oscillates, there are fluctuations in the track and direction of these depressions, and the intensity and the amount of rainfall vary from year to year.

9 The rain which comes in spells, displays a declining trend from west to east over the west coast, and from the southeast towards the northwest over the North Indian Plain and the northern part of the Peninsula.

THE NATURE OF INDIAN MONSOON

2 BREAK IN THE MONSOON

1. During the south-west monsoon period after having rains for a few days, if rain fails to occur for one or more weeks, it is known as break in the monsoon.
2. These dry spells are quite common during the rainy season.
3. These breaks in the different regions are due to different reasons:
   i. In northern India rains are likely to fail if the rain-bearing storms are not very frequent along the monsoon trough or the ITCZ over this region.
   ii. Over the west coast the dry spells are associated with days when winds blow parallel to the coast.

Learning Space

THE RHYTHM OF SEASONS

1 The climatic conditions of India can best be described in terms of an annual cycle of seasons.

2 The meteorologists recognise the following four seasons :

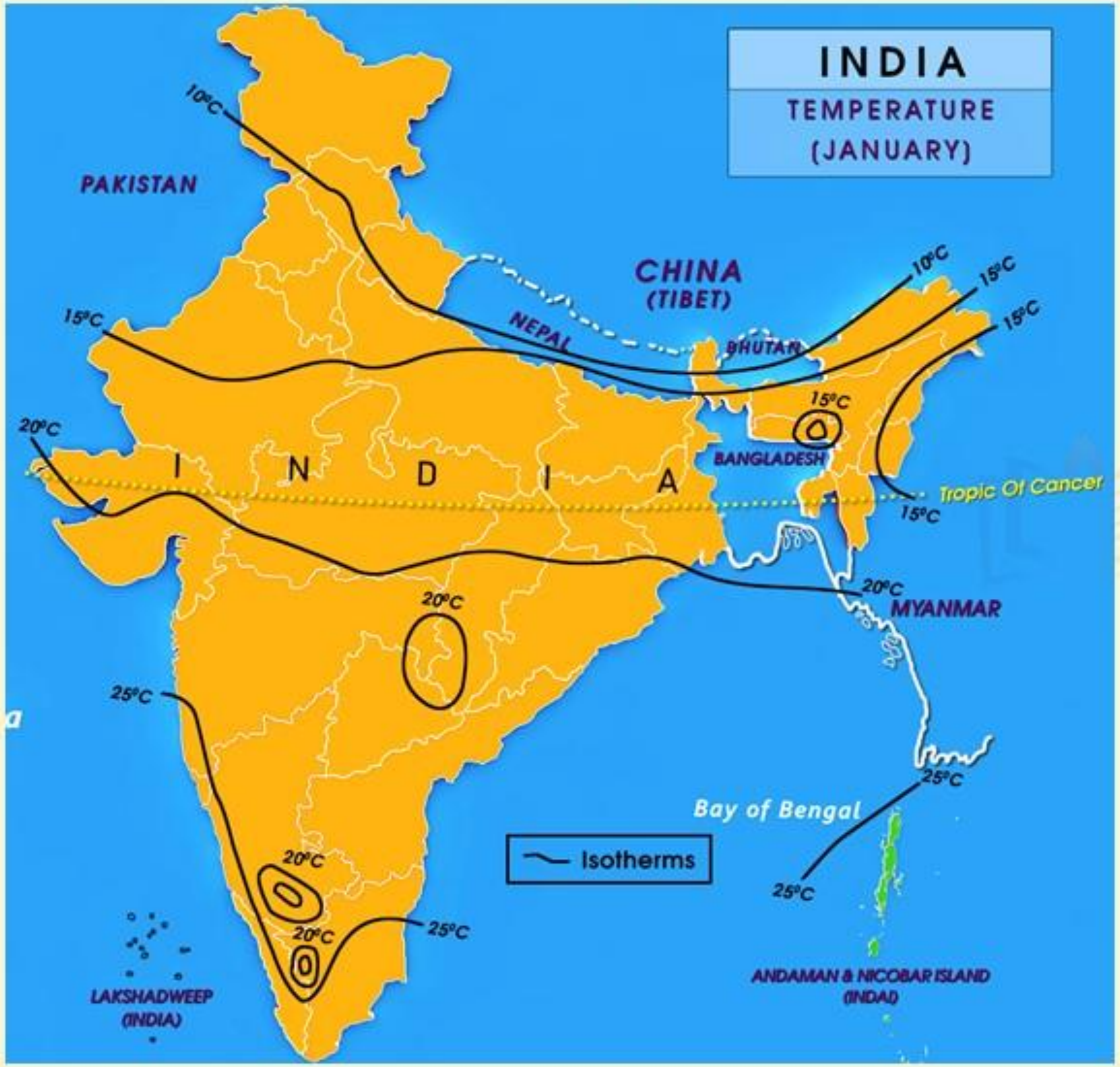
- the cold weather season
- the hot weather season
- the southwest monsoon season
- the retreating monsoon season.

Learning Space

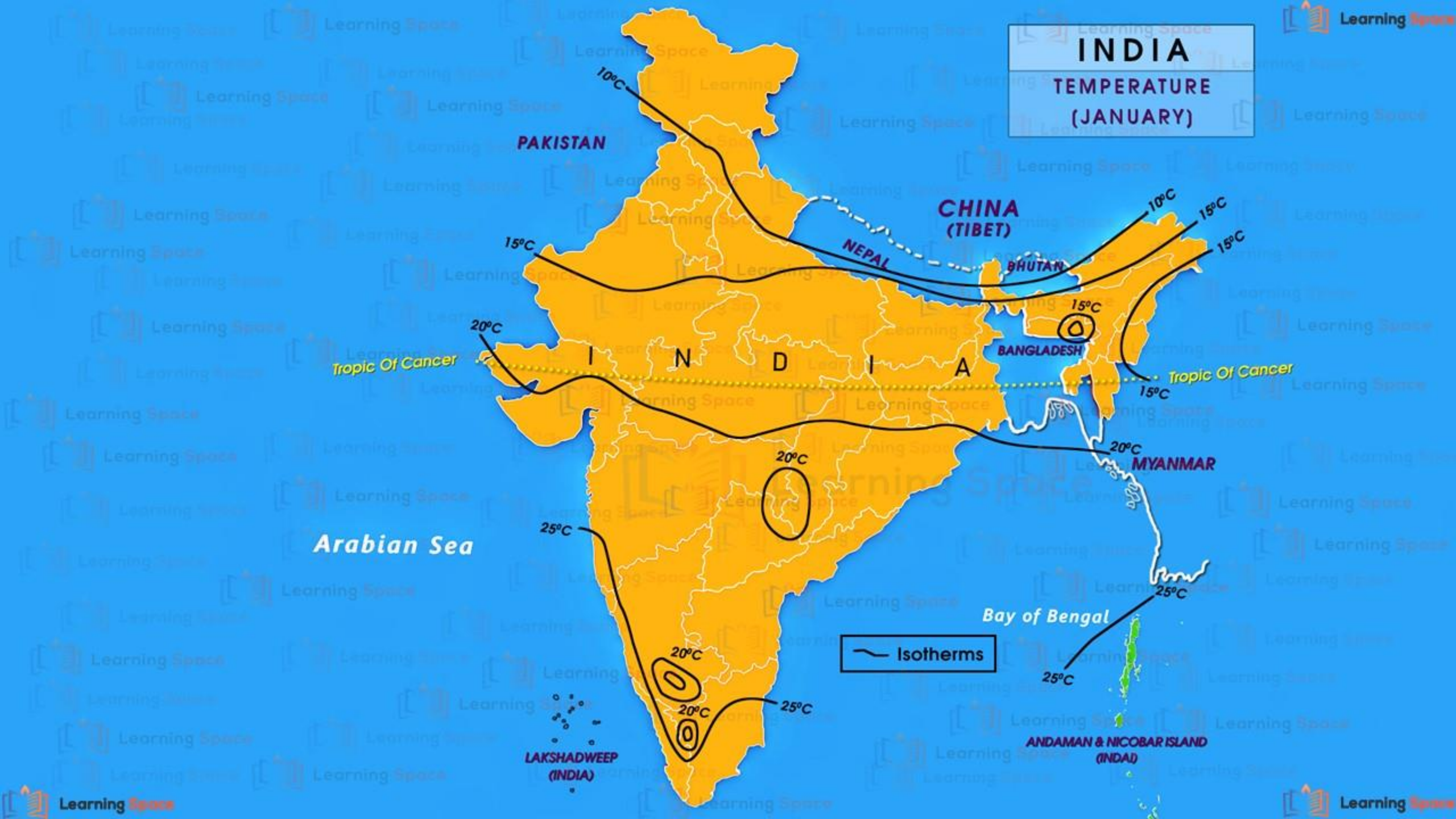
THE RHYTHM OF SEASONS

1 THE COLD WEATHER SEASON

TEMPERATURE



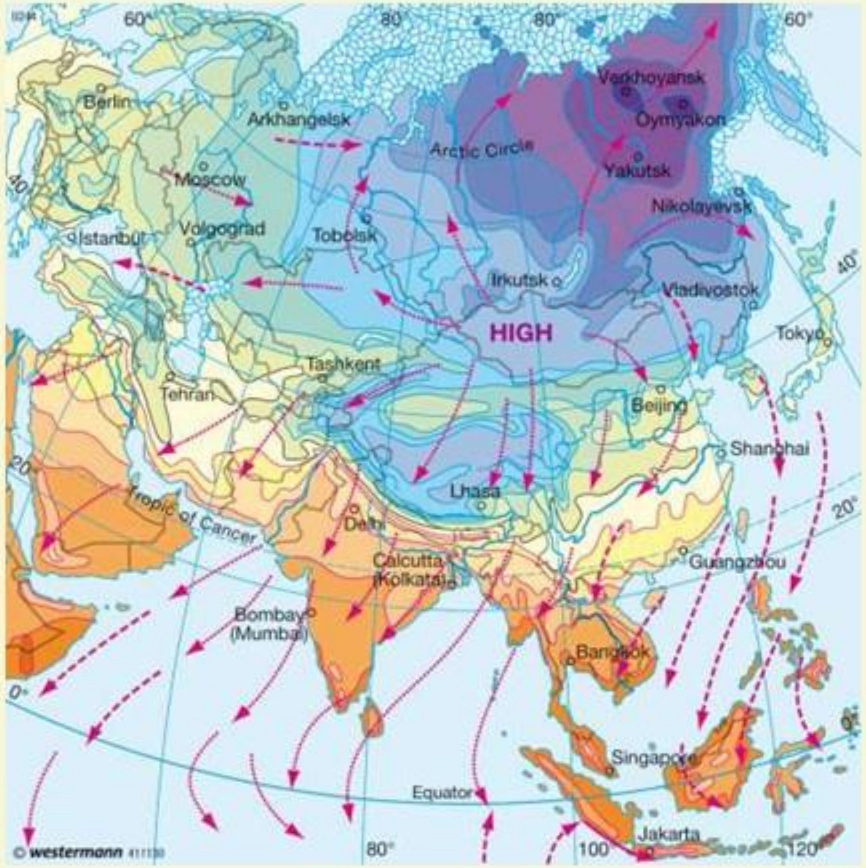
1. Usually, the cold weather season sets in by mid-November in northern India.
2. December and January are the coldest months in the northern plain.
3. The mean daily temperature remains below 21°C over most parts of northern India.
4. The night temperature may be quite low, sometimes going below freezing point in Punjab and Rajasthan.


INDIA
TEMPERATURE
(JANUARY)
PAKISTAN
CHINA
(TIBET)
NEPAL
BHUTAN
BANGLADESH
MYANMAR
I N D I A
Tropic Of Cancer
Tropic Of Cancer
Arabian Sea
Bay of Bengal
LAKSHADWEEP
(INDIA)
ANDAMAN & NICOBAR ISLAND
(INDAI)
Isotherms
10°C
15°C
20°C
25°C
Learning Space

THE RHYTHM OF SEASONS

1 THE COLD WEATHER SEASON

TEMPERATURE



5 There are three main reasons for the excessive cold in north India during this season :

i. States like Punjab, Haryana and Rajasthan being far away from the moderating influence of sea experience continental climate.

ii. The snowfall in the nearby Himalayas.

iii. Around February, the cold winds coming from the Caspian Sea and Turkmenistan bring cold wave along with frost and fog over the northwestern parts of India.

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THE RHYTHM OF SEASONS

1 THE COLD WEATHER SEASON

TEMPERATURE

6 The Peninsular region of India, however, does not have any well-defined cold weather season.

7 There is hardly any seasonal change in the distribution pattern of the temperature in coastal areas because of moderating influence of the sea and the proximity to equator.

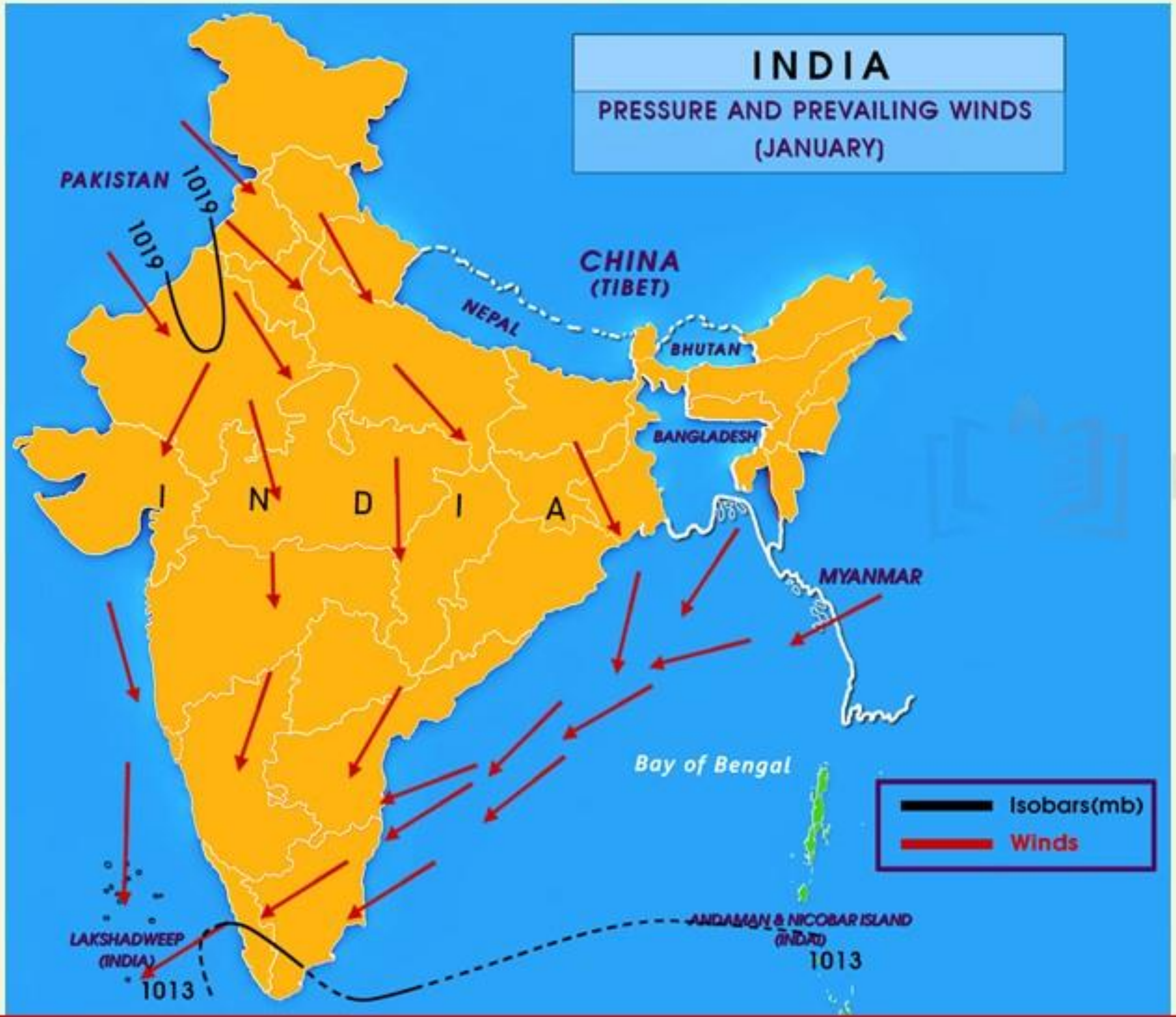
8 For example, the mean maximum temperature for January at Thiruvananthapuram is as high as 31°C, and for June, it is 29.5°C.

9 Temperatures at the hills of Western Ghats remain comparatively low.

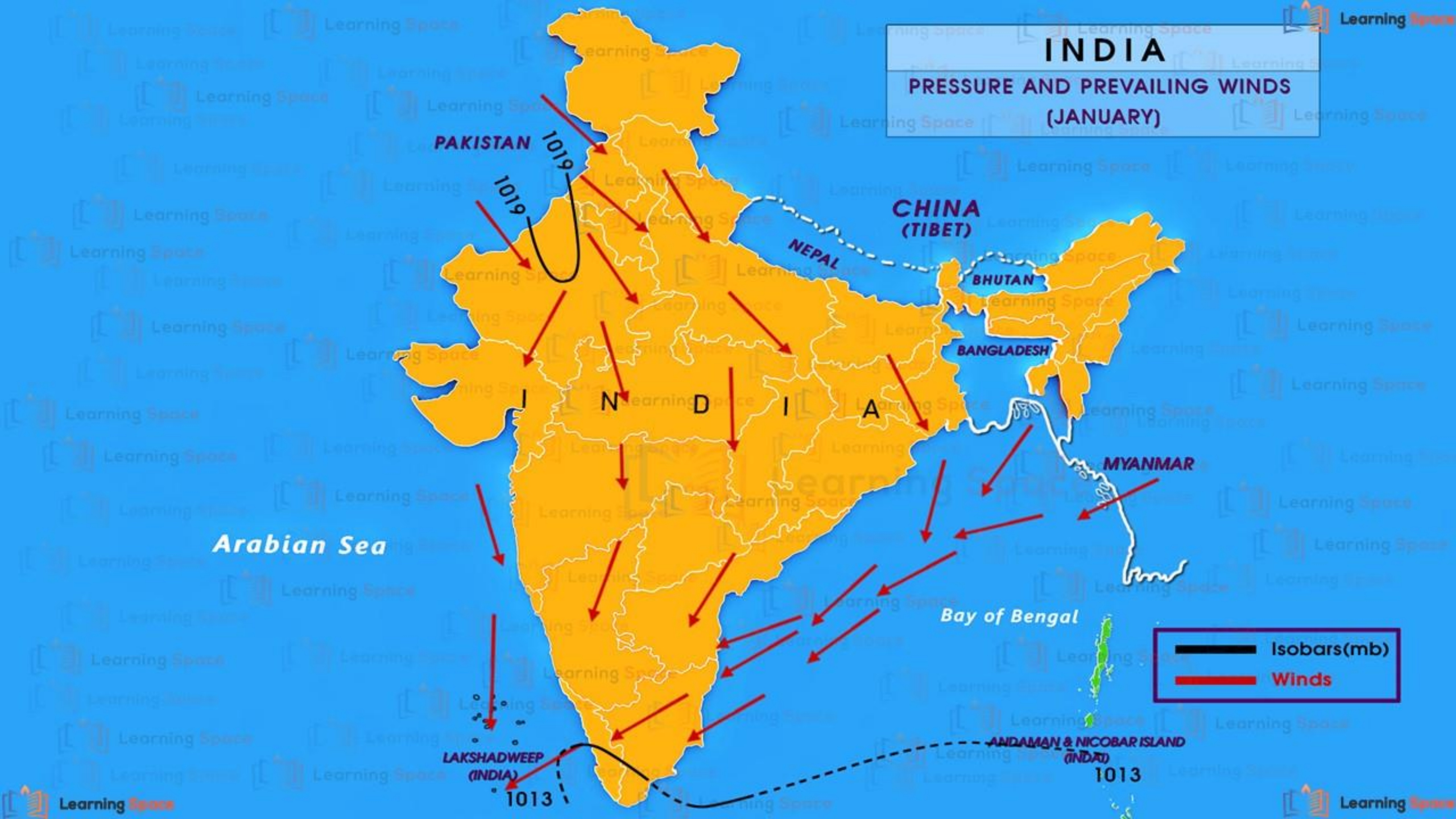
THE RHYTHM OF SEASONS

1 THE COLD WEATHER SEASON

PRESSURE AND WINDS



1. By the end of December 22nd December), the sun shines vertically over the Tropic of Capricorn in the southern hemisphere.
2. The weather in this season is characterised by feeble high pressure conditions over the northern plain.
3. In south India, the air pressure is slightly lower.
4. The isobars of 1019 mb and 1013 mb pass through northwest India and far south, respectively.



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THE RHYTHM OF SEASONS

1 THE COLD WEATHER SEASON

PRESSURE AND WINDS

5 As a result, winds start blowing from northwestern high pressure zone to the low air pressure zone over the Indian Ocean in the south.

6 Due to low pressure gradient, the light winds with a low velocity of about 3-5 km per hour begin to blow outwards.

7 By and large, the topography of the region influences the wind direction.

8 They are westerly or northwesterly down the Ganga Valley. They become northerly in the Ganga-Brahmaputra delta.

9 Free from the influence of topography, they are clearly northeasterly over the Bay of Bengal.

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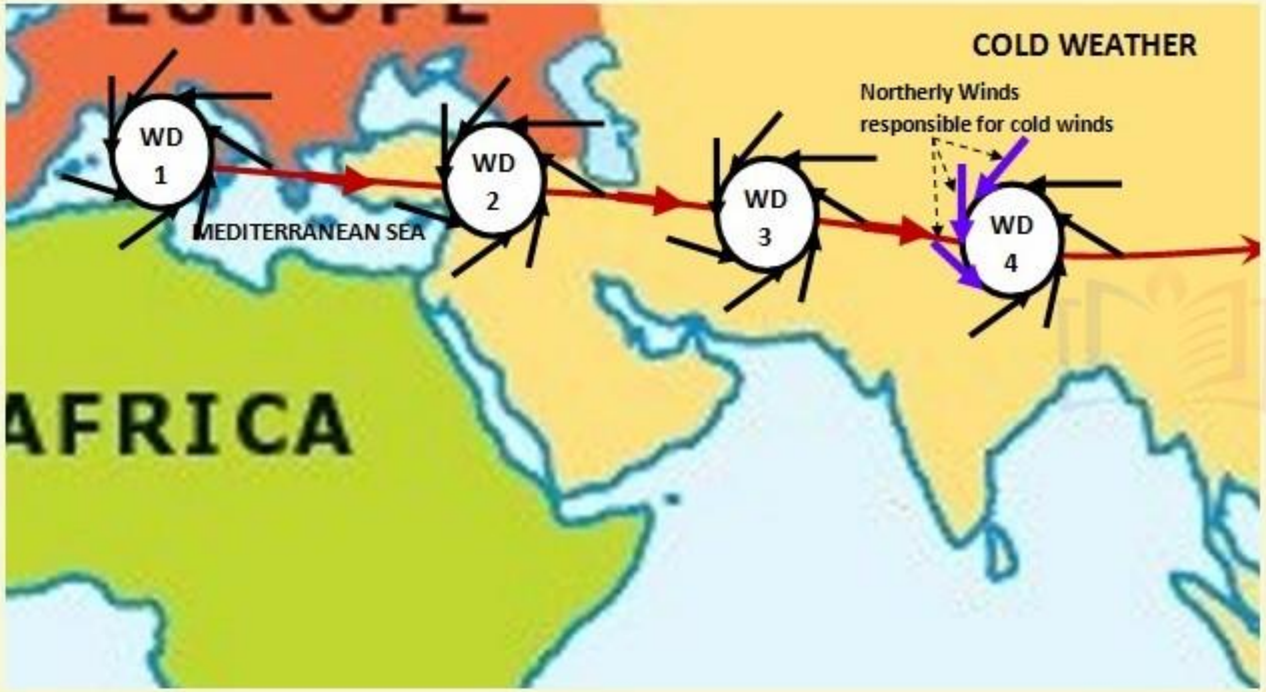
THE RHYTHM OF SEASONS

1 THE COLD WEATHER SEASON

PRESSURE AND WINDS



10 During the winters, the weather in India is pleasant. The pleasant weather conditions, however, at intervals, get disturbed by shallow cyclonic depressions originating over the east Mediterranean Sea and travelling eastwards across West Asia, Iran, Afghanistan and Pakistan before they reach the northwestern parts of India.

11 On their way, the moisture content gets augmented from the Caspian Sea in the north and the Persian Gulf in the south.

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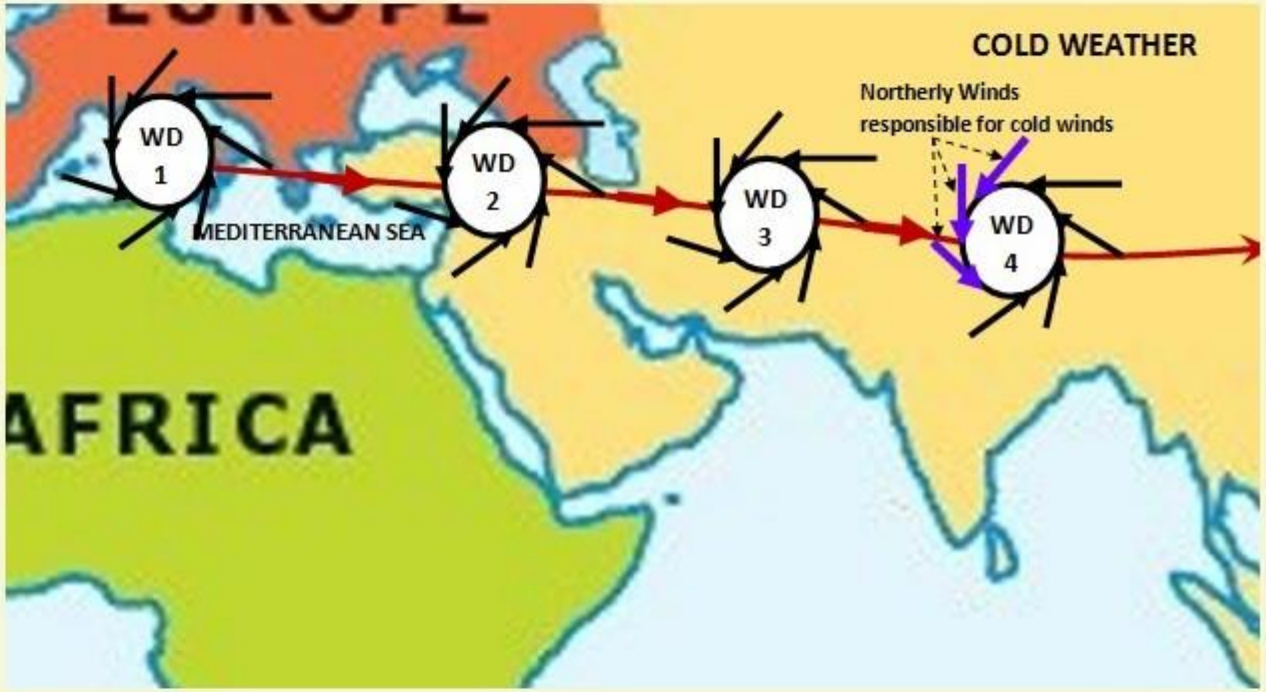
THE RHYTHM OF SEASONS

1 THE COLD WEATHER SEASON

PRECIPITATION



1. Winter monsoons do not cause rainfall as they move from land to the sea.

2. It is because firstly, they have little humidity and secondly, due to anti-cyclonic circulation on land, the possibility of rainfall from them reduces. So, most parts of India do not have rainfall in the winter season.

3. The passage of western disturbance causes widespread fog and cold waves lowering the minimum temperature by 5° to 10°C below normal are experienced. Fog lowers visibility and causes great inconvenience for transportation.

THE RHYTHM OF SEASONS

1 THE COLD WEATHER SEASON

RAINFALL

POINT NO. I

However, there are some exceptions to it:

1. In northwestern India, some weak temperate cyclones from the Mediterranean sea cause rainfall in Punjab, Haryana, Delhi and western Uttar Pradesh.
2. Although the amount is meagre, it is highly beneficial for rabi crops. The precipitation is in the form of snowfall in the lower Himalayas.
3. It is this snow that sustains the flow of water in the Himalayan rivers during the summer months.
4. The precipitation goes on decreasing from west to east in the plains and from north to south in the mountains.
5. The average winter rainfall in Delhi is around 53 mm. In Punjab and Bihar, rainfall remains between 25 mm and 18 mm respectively

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THE RHYTHM OF SEASONS

1 THE COLD WEATHER SEASON

RAINFALL

POINT NO. II

Central parts of India and northern parts of southern Peninsula also get winter rainfall occasionally.

POINT NO. III

Arunachal Pradesh and Assam in the northeastern parts of India also have rains between 25 mm and 50 mm during these winter months.

POINT NO. IV

During October and November, northeast monsoon while crossing over the Bay of Bengal, picks up moisture and causes torrential rainfall over the Tamil Nadu coast, southern Andhra Pradesh, southeast Karnataka and southeast Kerala.

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